

## Performance Testing

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|----------------------|-----------------|
| <b>Date</b>          | 30.09.2026      |
| <b>Team ID</b>       | SWTID-2026-7387 |
| <b>Project Name</b>  | FitBuddy        |
| <b>Maximum Marks</b> | 5 Marks         |

### Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

#### Step 1: Testing Overview

| Field                      | Details                                             |
|----------------------------|-----------------------------------------------------|
| <b>Testing Tool Used</b>   | Apache JMeter                                       |
| <b>Type of Testing</b>     | Load Testing and Stress Testing                     |
| <b>Target Module / API</b> | User Login, Workout Plan and Progress Tracking APIs |
| <b>Test Environment</b>    | Localhost / Development Server                      |
| <b>Test Date</b>           | 30.09.2026                                          |

#### Step 2: Test Scenarios

| S.No | Test Scenario / Description | No. of Virtual Users | Duration (sec) | Expected Outcome                      |
|------|-----------------------------|----------------------|----------------|---------------------------------------|
| 1    | Normal Load Test            | 50                   | 120            | Response time < 2 sec; no errors      |
| 2    | Peak Load Test              | 100                  | 180            | System remains stable with <1% errors |
| 3    | Stress Test                 | 200                  | 180            | Graceful handling under heavy load    |
| 4    | Spike Test                  | 300                  | 60             | System recovers without crash         |

### Step 3: Performance Test Results

| S.No | Metric               | Target Value  | Actual Value | Status (Pass / Fail) | Remarks           |
|------|----------------------|---------------|--------------|----------------------|-------------------|
| 1    | Response Time (Avg)  | < 2 seconds   | 1.2 sec      | Pass                 | Within target     |
| 2    | Response Time (Max)  | < 5 seconds   | 3.4 sec      | Pass                 | Within target     |
| 3    | Throughput (Req/sec) | >= 30 req/sec | 42 req/sec   | Pass                 | Stable throughput |
| 4    | Error Rate           | < 1%          | 0.4%         | Pass                 | Below 1%          |
| 5    | CPU Utilization      | < 80%         | 68%          | Pass                 | Below 80%         |
| 6    | Memory Utilization   | < 80%         | 72%          | Pass                 | Below 80%         |

### Step 4: Observations & Analysis

#### Key Findings:

FitBuddy maintained stable response times and low error rates under normal and peak loads.

#### Bottlenecks Identified:

Higher CPU and memory usage was observed during 200-300 virtual-user stress/spike tests.

#### Optimization Steps Taken:

Reduced unnecessary API calls, optimized database queries, and enabled response caching.

### Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Performance Test Evidence · JMeter

Execution Summary: 50-300 Virtual Users